

Education

Johns Hopkins University

Master's of Science in Engineering in Robotics

Baltimore, MD

August 2025 – June 2027

University of Toronto

Bachelor's of Applied Science in Mechanical Engineering & PEY Co-op

Toronto, Canada

September 2020 – June 2025

Project Experience

Autonomous Bleeding Control in Partial Nephrectomy

Baltimore, MD

IMERSE Lab, Johns Hopkins University

February 2026 – Present

- Fabricated patient-specific hydrogel kidney phantoms, designing 3D printed molds for silicone perfusion channels to simulate controlled intraoperative bleeding for robot-assisted partial nephrectomy.
- Developed a YOLO and U-Net++ vision pipeline for bleeding localization and segmentation using annotated dVRK endoscopic images and videos from 3D Slicer.
- Extending the platform to real-time bleeding source localization and classification for autonomous dVRK robotic control of suction and cautery assistance.

Visual Information through Sensory Tactile Array (VISTA)

Baltimore, MD

HAMR Lab, Johns Hopkins University

August 2025 – Present

- Designed a 7×5 tactile pin-array display to translate visual shape and color into height-encoded tactile representations for accessible museum exhibits, using a 3D printing cam system and laser cut housing.
- Developed a Python image-processing pipeline and Arduino closed-loop controller to actuate 35 motors for tactile rendering.
- Evaluated the system through preliminary user studies on tactile recognition of flag shapes and colors information; presented a Work-in-Progress paper and poster at the IEEE Haptics Symposium 2026.

Dual-Configuration Truck Tailgate Capstone Design Project

Toronto, ON

Lake Harbour & University of Toronto

September 2024 – April 2025

- Designed a retrofit guide-rail and dual-post actuation mechanism for an F-150 tailgate to incorporate vertical and rotational motion for direct bumper access.
- Led machine shop fabrication activities and coordinated drill press operations while ensuring safe machining practices and dimensional accuracy.
- Collaborated with industry client Lake Harbour over an 8-month development cycle, delivering progress reports, engineering documentation, and a final prototype showcased at the Capstone Design Symposium.

Autonomous Rover Design Project

Toronto, Canada

University of Toronto

September 2024 – December 2024

- Designed an autonomous rover platform integrating Bluetooth motor control, ultrasonic sensing, and servo actuation to study navigation, reorientation, obstacle detection and route planning with object pickup.
- Implemented Python-based maze navigation algorithms and validated system behavior through simulation-based testing, achieving consistent full-task completion along specified path within a 5-minute operation window.
- Designed a power distribution network using buck converters to supply stable 8V and 5V rails for motors and onboard electronics during continuous operation.

Window Cable Tensioner Redesign

Toronto, Canada

Magna International & University of Toronto

September 2023 – December 2023

- Redesigned a frameless car-door cable tensioner by identifying failure modes of fatigue, wear and stress concentrations in plastic components and spring mechanism, while preserving one-way cable tensioning functionality.
- Leveraged SolidWorks and ANSYS Mechanical to iterate 4 design variants, minimizing mechanical complexity, validating structural performance, and reducing manufacturing cost.
- Received the highest team rating in the course, with mentor feedback citing the design as most accurately meeting design requirements proven by FEA results.

Work Experience

Systems Engineer Internship

Toronto, Canada

WSP Canada

May 2023 – May 2024

- Authored a novel methodology for characterizing thermal properties of train fire spread using material testing data, informing CFD boundary conditions and adopted across the Calgary Green Line project.
- Performed hydraulic modeling and fire line design in AFT Fathom 13 to ensure compliance with NFPA 130 and NFPA 14 volumetric flow and pressure constraints.
- Supported RFP submissions by drafting technical sections and compiling proposal documentation, and maintained document control by tracking deliverable revisions and client comments across multidisciplinary teams.

Technical Skills

Software & Analysis: SolidWorks, ANSYS Mechanical and Fluent, OnShape

Programming: Python, MATLAB, Arduino IDE, C, C++, ROS2